



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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1. PRODUCT OVERVIEW

1.1 Product Name

KWALLO — from the Hausa word for football. Pronounced: Kwal-lo.

Domain suggestion: kwallo.ng or kwallo.com.ng

1.2 Product Vision

Kwallo is a Nigerian football platform built for the Northern Nigerian and Hausa-speaking audience. It shows live scores, generates match analysis using AI, runs a prediction game, and lets users download shareable match cards. Everything works in both English and Hausa.

1.3 The Problem We Are Solving

Problem	Current Reality	Kwallo Solution
No Hausa football content	Zero major platform covers football in Hausa	Full Hausa/English toggle on all content
NPFL ignored	Local league almost invisible online	NPFL scores and news covered daily
High data usage	Most sites load heavy images and scripts	Data-light mode for low data users
No shareable cards	Fans cannot share match results beautifully	Auto-generated downloadable image cards
No local predictions	No prediction game for Nigerian fans	Free prediction leaderboard — no betting
Generic AI summaries	No AI match analysis in any Nigerian app	AI previews and reports in English & Hausa

1.4 Target Audience

- Northern Nigerian football fans (primary)
- All Nigerian football fans (secondary)
- Super Eagles supporters globally (tertiary)
- NPFL fans and local league followers

1.5 Monetization Model

- Google AdSense — display ads on all pages
- Direct ad placements for Nigerian brands (future)
- Sponsored match cards and prediction rounds (future)
- Premium features for power users (future)

2. COMPLETE FEATURE SPECIFICATION

2.1 Core Features (MVP — Week 1)

2.1.1 Live Scores & Fixtures

- Real-time match scores updated every 60 seconds
- Fixtures: upcoming matches with Nigerian time (WAT) display
- Results: completed matches with full stats
- Competitions covered: EPL, UCL, La Liga, AFCON, Super Eagles, NPFL
- Match detail page: lineups, scorers, timeline, stats
- Match status badges: LIVE (red), HT, FT, Postponed

2.1.2 Standings & Tables

- League tables for all covered competitions
- Form guide (last 5 matches) for each team
- Top scorers list
- Team profile page: squad, recent results, upcoming fixtures

2.1.3 NPFL Coverage (Nigerian Professional Football League)

- NPFL scores and standings — daily updates
- NPFL match previews (AI-generated)
- NPFL top scorers and team stats
- No major platform covers NPFL well — this is the clearest gap in the market

2.1.4 AI-Generated Content

- Pre-match preview: team form, head-to-head, key players, prediction
- Post-match report: goals, performance analysis, ratings
- Transfer news summaries in simple English and Hausa
- All AI content generated automatically from API data — no human writer needed
- Content refreshes before every match and after every final whistle

2.2 Shareable Features

2.2.1 Downloadable Match Cards

Users download image cards and share them on WhatsApp. Every card carries the Kwallo name. That is how the platform spreads without spending money on ads.

Card Types:

- PRE-MATCH HYPE CARD: Team logos facing each other, match date in WAT, venue, competition badge
- FINAL SCORECARD: Team names, score, goal scorers, match stats, man of the match
- I CALLED IT CARD: User's prediction vs actual result — "I predicted this on Kwallo" — shareable pride card
- STANDINGS SNAPSHOT: Top 5 teams in any league, auto-updated, one-click download
- HEAD-TO-HEAD CARD: Last 5 meetings between two teams before a big match
- SUPER EAGLES JOURNEY CARD: Tournament bracket progress for national team
- PLAYER OF THE WEEK CARD: Best performer stats card

Technical implementation:

- Python Pillow library generates all cards on the backend
- Cards generated from live API data — no manual design needed
- Consistent branding: Kwallo logo, green color scheme, clean minimal design
- Format: 1080x1080px (perfect for WhatsApp status and Instagram)
- Download endpoint returns PNG file directly

2.2.2 WhatsApp Sharing

- Every page has a WhatsApp share button
- Shared links show rich preview (Open Graph tags) with match image
- Share message auto-formats: match, score, and Kwallo link
- Cards are pre-sized for WhatsApp status (1080x1080)

2.3 Community Features

2.3.1 Prediction Game

- Users predict match score before kickoff
- Points system: exact score = 3pts, correct winner = 1pt
- NO MONEY — pure bragging rights and leaderboard
- Monthly leaderboard resets — keeps competition fresh
- Prediction locks when match kicks off
- After match: instant result with points earned

- Streak tracker: how many correct predictions in a row
- "I Called It" card auto-generates for correct predictions

2.3.2 Match Debates

- Comment section on every match page
- Pre-match poll: Who will win? (quick one-tap vote)
- Post-match poll: Rate this match 1-5 stars
- Hot comment pinned at top (most liked)
- Simple upvote system — no downvotes (keep it positive)

2.3.3 Hot Takes Section

- Users post short football opinions (280 characters max)
- Others vote: AGREE or DISAGREE
- Most controversial takes featured on homepage
- Filter by team or competition

2.3.4 Fan Profile & Badges

- User picks their favourite club
- Badge displayed on all their activity
- Simple profile: prediction record, favourite team, streak
- No complex social features — keep it focused

2.4 Nigeria-First Features

2.4.1 Hausa Language Toggle

- Every page available in both English and Hausa
- User sets preference — remembered in local storage
- AI content generated in both languages simultaneously
- Match card downloads available in both languages
- URL structure: /en/ and /ha/ prefixes

2.4.2 Nigerian Time (WAT) Display

- All fixture times shown in West Africa Time automatically
- Countdown timer to next match in WAT
- No UTC confusion — Nigerians always see their local time

2.4.3 Data-Light Mode

- Toggle button in header — one click activation
- Removes all images, loads only text and scores
- Reduces page size by ~80%
- Remembers user preference
- This alone keeps users coming back when data is limited

2.4.4 PWA (Progressive Web App)

- Users can install Kwallo like an app on Android
- Works offline — cached scores available without internet
- Push notifications for match start and final whistle (future)
- Fast load time even on 2G/3G networks

3. FUNCTIONAL REQUIREMENTS

3.1 User Authentication

ID	Requirement	Priority
FR-001	Users can register with email and password	High
FR-002	Users can log in and log out	High
FR-003	JWT-based authentication with HttpOnly cookies	High
FR-004	Password reset via email	Medium
FR-005	Guest users can view all content (no login required)	High
FR-006	Login required only for predictions, comments, and hot takes	High
FR-007	User profile page with prediction history	Medium

3.2 Scores & Fixtures

ID	Requirement	Priority
FR-010	Display live scores with auto-refresh every 60 seconds	High
FR-011	Show fixtures for next 7 days	High
FR-012	Show results for last 7 days	High
FR-013	Match detail page with full stats	High
FR-014	Filter matches by competition	High
FR-015	Display all times in WAT (West Africa Time)	High
FR-016	NPFL scores and fixtures fully covered	High
FR-017	Standing tables for all competitions	High
FR-018	Top scorers list per competition	Medium

3.3 Image Card Generation

ID	Requirement	Priority
FR-020	Generate pre-match hype card for every fixture	High

FR-021	Generate final scorecard after match ends	High
FR-022	Generate I-Called-It card for correct predictions	High
FR-023	Generate standings snapshot card (top 5)	Medium
FR-024	Generate head-to-head history card	Medium
FR-025	Cards available in English and Hausa	High
FR-026	All cards output as PNG 1080x1080px	High
FR-027	Kwallo logo and branding on every card	High
FR-028	Download button visible on all match pages	High
FR-029	Card generation takes less than 3 seconds	High

3.4 Prediction System

ID	Requirement	Priority
FR-030	Authenticated users submit score predictions before kickoff	High
FR-031	Predictions lock at match kickoff time	High
FR-032	Auto-score predictions after match ends	High
FR-033	Monthly leaderboard with top predictors	High
FR-034	User streak tracker on profile	Medium
FR-035	I-Called-It card triggered for exact predictions	High
FR-036	No money, no betting — points and ranks only	High

3.5 AI Content

ID	Requirement	Priority
FR-040	AI generates match preview 24hrs before kickoff	High
FR-041	AI generates post-match report within 1hr of final whistle	High
FR-042	Content generated in both English and Hausa	High
FR-043	AI content saved to database (not regenerated each load)	High
FR-044	Transfer news summarized daily	Medium
FR-045	AI content clearly labeled as AI-generated	High

3.6 Community

ID	Requirement	Priority
FR-050	Comments on match pages (login required)	High
FR-051	Pre-match poll: Who will win?	High
FR-052	Post-match poll: Rate this match	Medium
FR-053	Hot Takes section with agree/disagree voting	Medium
FR-054	User picks favourite club on registration	Medium
FR-055	Fan badge displayed on profile and comments	Medium

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance

- Homepage loads in under 2 seconds on 3G connection
- Data-light mode loads in under 1 second
- API response time: under 500ms for score endpoints
- Image card generation: under 3 seconds
- System handles 500 concurrent users on basic hosting

4.2 Scalability

- Django architecture with Celery for background tasks (AI generation, card creation)
- Redis for caching live scores — reduces API calls and speeds up responses
- Football API data cached for 60 seconds — not called on every request
- Image cards cached after first generation — not regenerated every download
- PostgreSQL with proper indexing for fast queries
- Stateless backend — easy to scale horizontally when needed

4.3 Reliability

- Football API fallback: if primary API fails, secondary API is called
- AI content fallback: if AI generation fails, last saved content is shown
- Error pages that still show scores even if one component fails
- Celery task retry on failure

4.4 Security

- JWT authentication stored in HttpOnly cookies (not localStorage)
- CSRF protection enabled
- Rate limiting on prediction and comment endpoints
- Input validation and sanitization on all user inputs
- Django security settings: HTTPS only, secure headers
- No sensitive data stored in frontend

4.5 Accessibility & UX

- Mobile-first design — 80% of users are on mobile

- Data-light mode for low data users
- Hausa language option for Northern Nigerian users
- All times in WAT — no UTC confusion
- Minimal clicks to reach live scores (max 1 click from homepage)
- PWA installable for app-like experience

4.6 SEO

- Server-side rendering for all score and news pages (good for Google)
- Open Graph tags for WhatsApp/social sharing previews
- Structured data markup for match scores (Google rich results)
- Fast page speed (Google ranks fast sites higher)
- Hausa-language content = low competition keywords

5. SYSTEM ARCHITECTURE

5.1 Overview

Kwallo separates the frontend and backend cleanly. Lovable AI handles the frontend. Django handles the backend. They talk to each other through a REST API.

5.2 Tech Stack

Layer	Technology	Purpose
Frontend	Lovable AI (React)	UI — all pages users see
Backend	Django 4.2 + DRF	API, business logic, data
Database	PostgreSQL	Primary data storage
Cache	Redis	Live scores cache, session storage
Task Queue	Celery + Redis	AI content generation, card creation
AI Engine	FreeModel / Gemini API	Match previews and reports
Football Data	football-data.org + API-Football	Scores, fixtures, standings
Image Gen	Python Pillow	Downloadable match cards
Payments (ads)	Google AdSense	Monetization
Deployment	Railway or Render (free)	Hosting — zero cost to start
Storage	Cloudflare R2 or local	Generated card images

5.3 Django Backend Structure

App	Responsibility	Key Models
core	Base config, utils, health check	SiteSettings
accounts	User registration, login, profile	User, UserProfile
fixtures	Scores, results, standings (from API)	Match, Team, Competition, Standing

news	AI-generated content storage	MatchPreview, MatchReport, TransferNews
predictions	User predictions and scoring	Prediction, Leaderboard, UserStreak
cards	Image card generation and storage	MatchCard, CardTemplate
community	Comments, polls, hot takes	Comment, Poll, HotTake, Vote
notifications	Email and push alerts	Notification, NotificationPreference

5.4 API Endpoint Plan

Scores & Fixtures

Method	Endpoint	Description
GET	/api/v1/fixtures/live/	All live matches right now
GET	/api/v1/fixtures/today/	Today's fixtures
GET	/api/v1/fixtures/upcoming/	Next 7 days fixtures
GET	/api/v1/fixtures/results/	Last 7 days results
GET	/api/v1/fixtures/:id/	Single match detail + stats
GET	/api/v1/standings/:competition_id/	League table
GET	/api/v1/competitions/	All supported competitions
GET	/api/v1/teams/:id/	Team profile + recent form

AI Content

Method	Endpoint	Description
GET	/api/v1/news/preview/:match_id/	AI match preview
GET	/api/v1/news/report/:match_id/	AI post-match report
GET	/api/v1/news/transfers/	Latest transfer summaries

Image Cards

Method	Endpoint	Description
GET	/api/v1/cards/prematch/:match_id/	Download pre-match hype card
GET	/api/v1/cards/scorecard/:match_id/	Download final scorecard
GET	/api/v1/cards/standings/:competition_id/	Download standings card
GET	/api/v1/cards/h2h/:match_id/	Download head-to-head card
GET	/api/v1/cards/called-it/:prediction_id/	Download I-Called-It card

Predictions

Method	Endpoint	Description
POST	/api/v1/predictions/	Submit a prediction
GET	/api/v1/predictions/my/	User's own predictions
GET	/api/v1/predictions/leaderboard/	Monthly leaderboard
GET	/api/v1/predictions/streak/	User's current streak

Community

Method	Endpoint	Description
GET	/api/v1/comments/:match_id/	Get match comments
POST	/api/v1/comments/	Post a comment
GET	/api/v1/polls/:match_id/	Get match poll
POST	/api/v1/polls/:poll_id/vote/	Submit poll vote
GET	/api/v1/hottakes/	Get hot takes list
POST	/api/v1/hottakes/	Post a hot take
POST	/api/v1/hottakes/:id/vote/	Vote on a hot take

Accounts

Method	Endpoint	Description
POST	/api/v1/auth/register/	Create account
POST	/api/v1/auth/login/	Login — returns JWT cookie
POST	/api/v1/auth/logout/	Logout — clears cookie
GET	/api/v1/auth/profile/	Get user profile
PATCH	/api/v1/auth/profile/	Update profile (club, language)

6. DATABASE SCHEMA

6.1 Match Model

Field	Type	Description
id	UUID	Primary key
external_id	Integer	ID from football API
competition	FK → Competition	League or cup
home_team	FK → Team	Home team
away_team	FK → Team	Away team
kickoff_time	DateTime	Kickoff in UTC (display in WAT)
status	CharField	SCHEDULED / LIVE / HT / FT / POSTPONED
home_score	Integer	Home goals (null if not started)
away_score	Integer	Away goals (null if not started)
minute	Integer	Current match minute (if live)
venue	CharField	Match venue
last_updated	DateTime	Last API sync time

6.2 MatchCard Model

Field	Type	Description
id	UUID	Primary key
match	FK → Match	Related match
card_type	CharField	PREMATCH / SCORECARD / H2H / STANDINGS
language	CharField	EN or HA
image_file	FileField	Path to generated PNG
generated_at	DateTime	When the card was created
download_count	Integer	How many times downloaded

6.3 Prediction Model

Field	Type	Description
id	UUID	Primary key
user	FK → User	Who predicted
match	FK → Match	Which match
predicted_home	Integer	Predicted home score
predicted_away	Integer	Predicted away score
points_earned	Integer	0, 1 or 3 — set after match
is_scored	Boolean	Whether points have been awarded
created_at	DateTime	When prediction was submitted

6.4 MatchContent (AI) Model

Field	Type	Description
id	UUID	Primary key
match	FK → Match	Related match
content_type	CharField	PREVIEW or REPORT
content_en	TextField	English AI content
content_ha	TextField	Hausa AI content
generated_at	DateTime	When AI generated this
ai_model_used	CharField	Model name for tracking

6.5 UserProfile Model

Field	Type	Description
user	OneToOne → User	Django user
favourite_club	FK → Team	User's club
language_preference	CharField	EN or HA
prediction_points	Integer	Total points this month

prediction_streak	Integer	Current correct prediction streak
total_predictions	Integer	All-time prediction count

7. FRONTEND SPECIFICATION (FOR LOVABLE AI)

Give this section to Lovable AI as your prompt. It covers every page and how each one should look.

7.1 Design System

Element	Value
Primary Color	#1A7F4B (Deep Green)
Secondary Color	#1A1A2E (Dark Navy)
Accent	#F5A623 (Gold — for live badges and highlights)
Background	#F8F9FA (Light Gray)
Card Background	#FFFFFF
Font	Inter or DM Sans — clean and modern
Border Radius	12px on cards, 8px on buttons
Mobile First	All designs start at 375px width
Data-light Mode	Toggle removes all images, white background, text only

7.2 Pages to Build

Page	Route	What It Shows
Homepage	/	Live scores, today's fixtures, trending hot take, top prediction
Live Scores	/live	All live matches in real-time — auto refresh
Fixtures	/fixtures	Upcoming matches — filterable by competition
Results	/results	Completed matches — filterable by competition
Match Detail	/match/:id	Full match page: stats, AI preview/report, comments, cards, poll, predictions
Standings	/standings	League tables — tab per competition
NPFL Hub	/npfl	Dedicated NPFL page: scores, table, top scorers
Super Eagles	/super-eagles	National team fixtures, results, squad

Predictions	/predictions	Submit predictions, see leaderboard, my history
Leaderboard	/leaderboard	Monthly top predictors
Hot Takes	/hot-takes	Community opinions — vote agree/disagree
Profile	/profile	User stats, badges, prediction history, settings
Login	/login	Simple login form
Register	/register	Registration with club selection

7.3 Key UI Components

Match Card Component

- Shows: team names + logos, score (or kickoff time), competition badge, status badge
- LIVE badge in red/gold with pulsing dot animation
- Tap anywhere on card → goes to Match Detail page
- Download button on the card → triggers card image download
- Share button → opens WhatsApp share with pre-filled message
- Must be clean, minimal, fast to load

Navigation

- Bottom navigation bar on mobile: Home | Live | Predictions | Profile
- Top header: Kwallo logo + Language toggle (EN | HA) + Data-light toggle
- Competition filter tabs: EPL | UCL | NPFL | Super Eagles | La Liga

Download Card Button

- Green button labeled: Download Match Card
- On click: calls backend /api/v1/cards/... endpoint
- Shows loading spinner while generating
- Auto-downloads PNG file to user's phone
- Shows small success toast: Card downloaded. Share on WhatsApp!

Language Toggle

- Simple EN / HA switch in header
- On toggle: all text content switches language instantly
- Preference saved in localStorage
- Match AI content fetches correct language from API

8. BACKEND BUILD GUIDE (DJANGO)

8.1 Project Setup

Command	Purpose
<code>django-admin startproject kwallo_backend</code>	Create project
<code>python manage.py startapp core</code>	Base app
<code>python manage.py startapp accounts</code>	Auth app
<code>python manage.py startapp fixtures</code>	Scores app
<code>python manage.py startapp news</code>	AI content app
<code>python manage.py startapp predictions</code>	Predictions app
<code>python manage.py startapp cards</code>	Image card app
<code>python manage.py startapp community</code>	Comments and polls app

8.2 Required Packages

Package	Purpose
<code>django</code>	REST API
<code>django-simplejwt</code>	JWT Auth
<code>django-cors-headers</code>	Allow Lovable frontend to call the API
<code>celery</code>	Background tasks (AI generation, card creation)
<code>redis</code>	Cache + Celery broker
<code>Pillow</code>	Image card generation
<code>requests</code>	Call football APIs
<code>python-decouple</code>	Environment variables (.env)
<code>gunicorn</code>	Production server
<code>whitenoise</code>	Serve static files
<code>psycopg2-binary</code>	PostgreSQL connector

8.3 Key Background Tasks (Celery)

Task	Schedule	What It Does
sync_live_scores	Every 60 seconds	Fetch live scores from API, update DB, update Redis cache
sync_fixtures	Every 6 hours	Fetch upcoming fixtures for next 7 days
generate_match_previews	Daily at 8AM	AI generates previews for all matches today
generate_match_reports	After match FT	AI generates post-match report
score_predictions	After match FT	Calculate and award points to users
pregenerate_cards	Daily at 7AM	Pre-generate match cards for today's fixtures
sync_standings	Every 24 hours	Update league tables

8.4 Image Card Generation Logic

In your cards app, create a generator.py file. The logic works like this:

- 1. Receive match_id and card_type
- 2. Fetch match data from database
- 3. Load base template image (1080x1080px green background)
- 4. Draw team names, logos, score using Pillow
- 5. Add Kwallo logo watermark at bottom
- 6. Save PNG to storage
- 7. Return file path for download endpoint

Card templates are written in Python. You set x,y coordinates for where each element goes. No design software needed.

8.5 Football API Integration

API	URL	Free Tier	Best For
football-data.org	api.football-data.org	10 calls/min, EPL/UCL	European leagues, international
API-Football (RapidAPI)	rapidapi.com/api-sports	100 calls/day	NPFL, more competitions

Use football-data.org as the main source. Pull NPFL data from API-Football. Cache every response in Redis for 60 seconds to stay within the free tier limits.

9. 14-DAY BUILD PLAN

Week 1 — Backend & Core Engine

Day	Task	Output
Day 1	Project setup, football API connected, live scores working	Scores visible in Django admin
Day 2	Fixtures, results, standings endpoints live	All score API endpoints responding
Day 3	NPFL coverage added via API-Football	NPFL scores working
Day 4	AI match preview and report generation	AI content saving to database
Day 5	Image card generation system (Pillow)	Cards downloading as PNG
Day 6	Predictions system: submit, score, leaderboard	Prediction API working
Day 7	User auth (register, login, JWT cookies)	Auth endpoints working

Week 2 — Community, Frontend Handoff & Launch

Day	Task	Output
Day 8	Comments, polls, hot takes endpoints	Community API working
Day 9	Celery tasks: auto-sync scores, auto-generate AI content	Background automation working
Day 10	API documentation (for Lovable AI frontend)	Clean API docs / Postman collection
Day 11	Deploy Django on Railway (free tier)	Live backend URL
Day 12	Frontend built in Lovable AI using API docs	Working UI connected to backend
Day 13	Testing: all features working end-to-end	Bug fixes done

Day 14	Submit to AdSense + Share in WhatsApp groups	First traffic incoming
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10. MONETIZATION & GROWTH PLAN

10.1 AdSense Setup

1. Apply to Google AdSense at launch (needs ~20+ pages of content)
2. Place ads: top of homepage, between fixtures, match detail page
3. Avoid intrusive ads — user experience first
4. Focus on traffic volume — more visitors = more ad money

10.2 Revenue Estimate (Honest)

Timeline	Daily Visitors	Estimated Monthly (AdSense)
Month 1-2	0 - 100	₦0 — building audience
Month 3	200 - 500	₦3,000 - ₦10,000
Month 6	1,000 - 3,000	₦20,000 - ₦60,000
Month 12	5,000 - 15,000	₦100,000 - ₦300,000+

These are conservative estimates. The card sharing feature is your main source of organic traffic. Each card shared puts the Kwallo name in front of new users at zero cost.

10.3 Growth Strategy

- WEEK 1: Share in every Nigerian football WhatsApp group you can find
- WEEK 2: Post highlights and predictions on Nigerian Twitter/X
- WEEK 3: Encourage users to share their I-Called-It cards
- MONTH 2: Reach out to small football content creators for mentions
- MONTH 3: Add NPFL team fanbase pages — local clubs have loyal local fans
- ONGOING: Hausa language = untapped SEO in Northern Nigeria

11. LAUNCH CHECKLIST

Before Going Live

5. Football API accounts created (football-data.org + API-Football RapidAPI)
6. Django project created with all 7 apps
7. PostgreSQL database connected
8. Redis running for cache and Celery
9. All API endpoints tested with Postman
10. Celery tasks running (auto-score sync, AI generation)
11. Image card generation working and downloading
12. Auth endpoints working (register, login, logout)
13. Deployed to Railway with environment variables set
14. Lovable AI frontend built and connected to backend API
15. Language toggle working (EN/HA)
16. Data-light mode working
17. All times showing in WAT
18. NPFL coverage confirmed working
19. Google AdSense application submitted

Day of Launch

20. Share in at least 20 Nigerian football WhatsApp groups
21. Post on Nigerian Twitter with a card screenshot as preview
22. Ask friends to register and make predictions
23. Monitor for errors on Railway dashboard
24. Check football API is returning data correctly